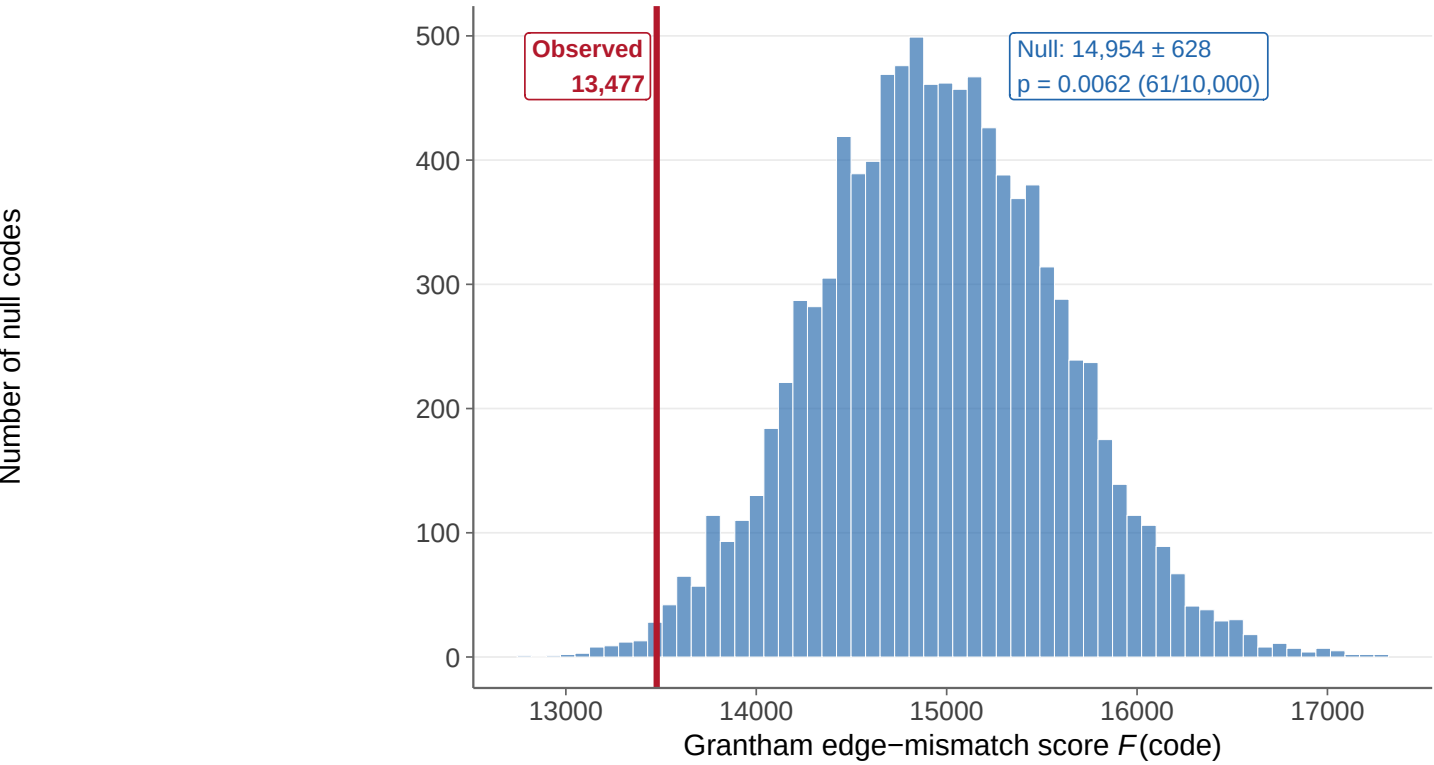


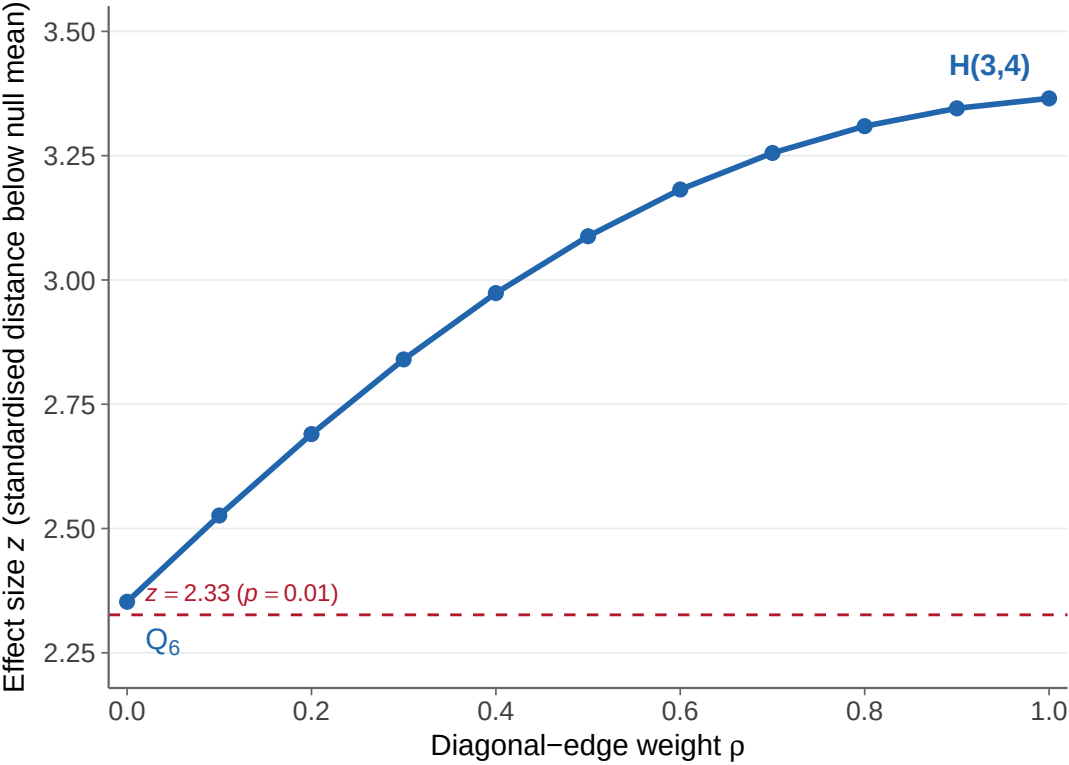
A
Hypercube Coloring Optimality

Standard code vs 10,000 quartet-pattern shuffle random colorings



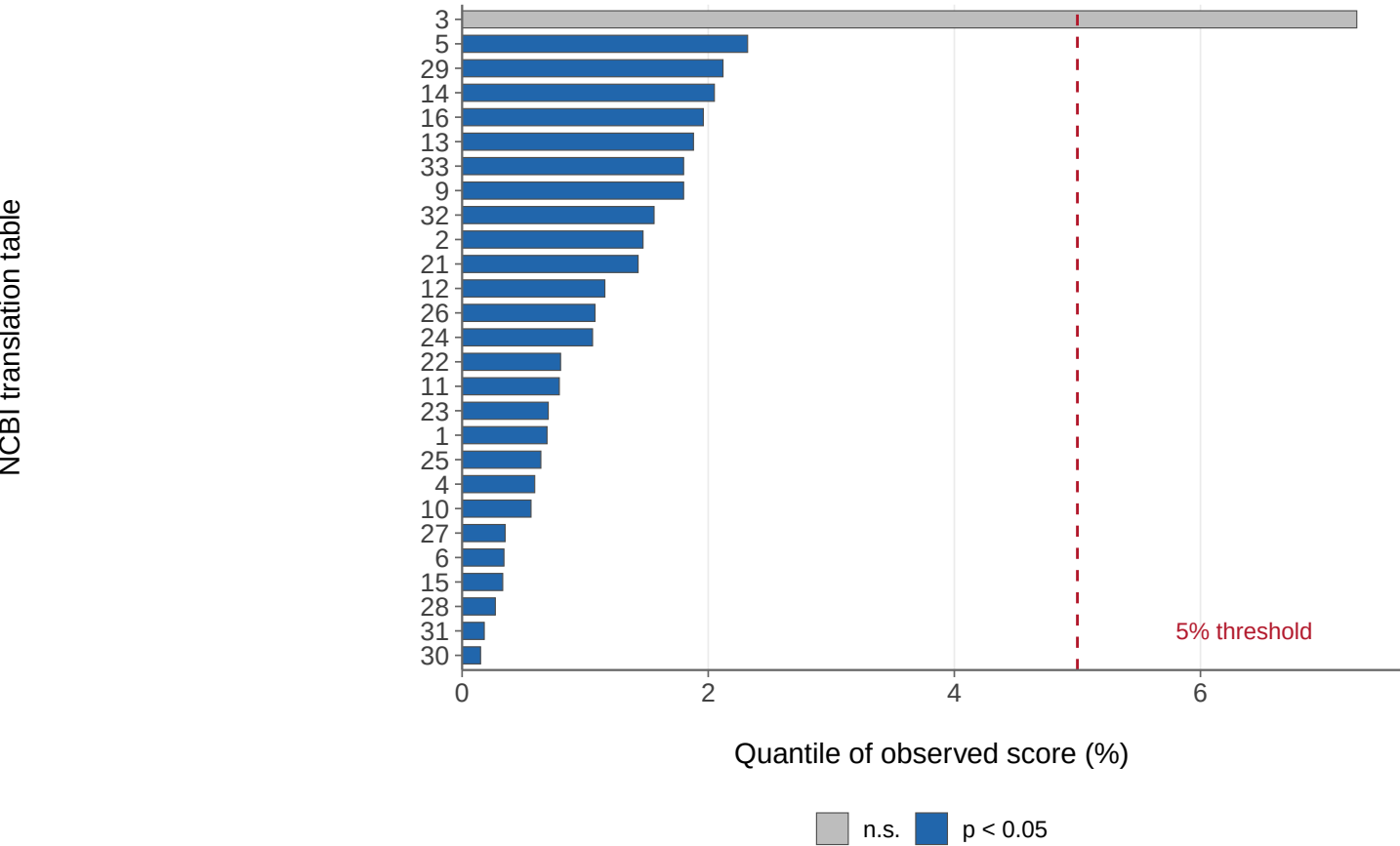
B
Optimality Robust Across Diagonal-Edge Weight ρ

$F_\rho = F_{H_1} + \rho \cdot F_{H_2}$; $\rho = 0$: Q_6 ; $\rho = 1$: $H(3, 4)$; $n = 10,000$



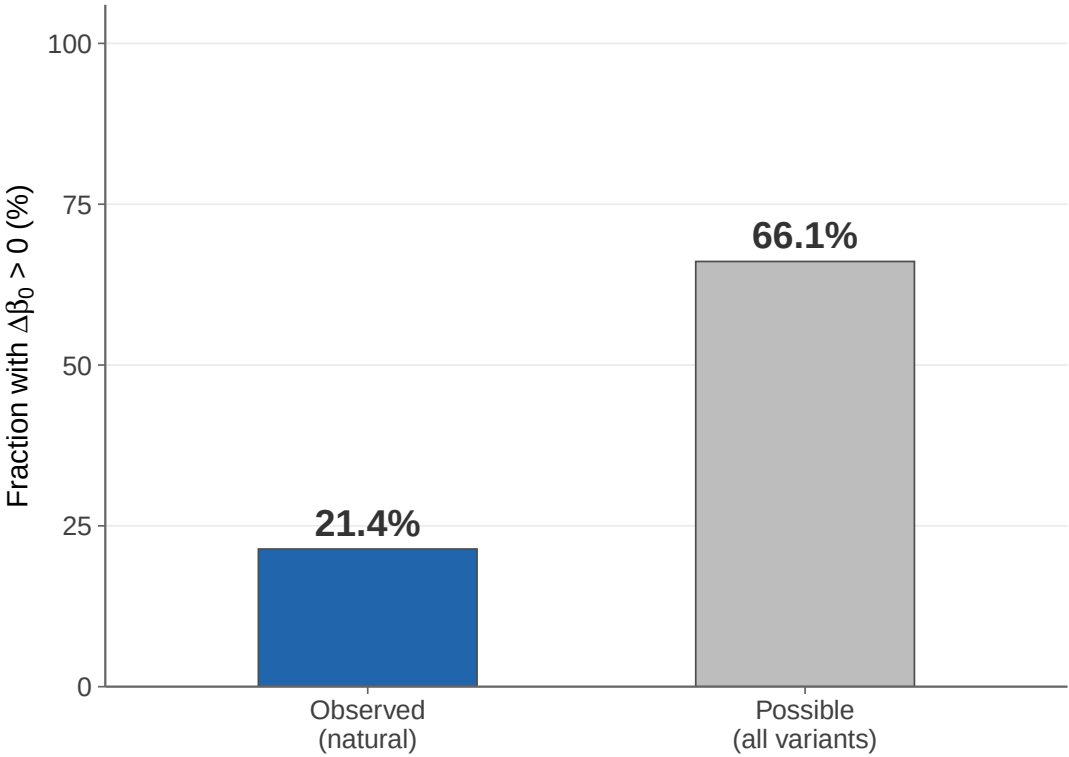
C
Coloring Optimality Preserved Across Variant Codes

Each NCBI table tested against its own block-preserving null ($n = 10,000$)



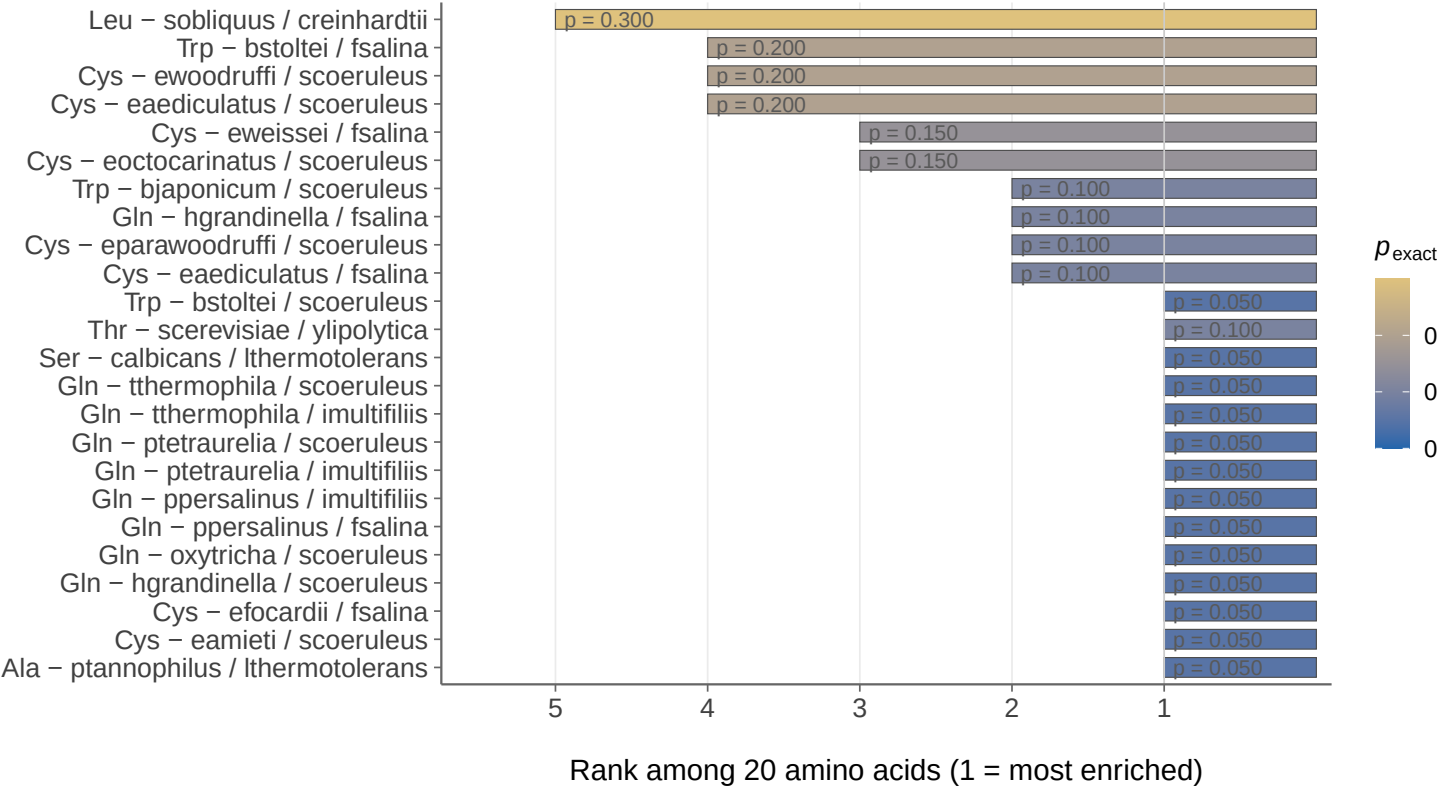
D
Natural Reassignments Avoid Topology-Breaking Changes

Observed 21.4% vs Possible 66.1% (RR = 0.32)
Hypergeom. $p = 1.3e-06$ | Permut. $p \leq 1e-04$ ($n = 10,000$)



E
Rank of Reassigned AA Among All tRNA Gene Counts

Rank 1 = most enriched amino acid in disconnection vs control comparison



F
Mismatch Score Decomposition by Codon Position

Total score: 13,477 -- position 2 dominates (amino acid identity changes)

